TASK 2: Build a Highly Available Web Application on AWS Using Linux, Bash, and Core AWS Services

TASK OVERVIEW

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1. Create a VPC with public/private subnets.

2. Launch EC2 instances in public subnets using a bash-script for user-data.

3. Store static content (images/configs) in S3 using the AWS CLI.

4. Attach EBS volume to EC2.

5. Mount EFS to EC2 for shared storage.

6. Set up IAM roles for EC2 & S3 access.

7. Create a Load Balancer (ELB).

8. Configure Auto Scaling Group (ASG).

9. Deploy RDS MySQL in private subnets.

10. Test high availability by scaling EC2 instances.

DETAILED TASK EXPLANATION

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Below is a step-by-step breakdown.

PART 1 — VPC Setup (Networking)

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Create a custom VPC:

✅ 1 VPC (CIDR: 10.0.0.0/16)

✅ Public Subnets (2) — for EC2 & Load Balancer

✅ Private Subnets (2) — for Auto Scaling + RDS

✅ Internet Gateway

✅ NAT Gateway

✅ Route Tables

PART 2 — IAM Role & Policies
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✅ An IAM Role for EC2 that allows:

Access to S3

Access to CloudWatch logs

Access to SSM (optional)

✅ An IAM user with only S3/EC2 permissions.

PART 3 — EC2 Setup with Bash User-Data Script
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Launch an EC2 instance in a public subnet.

Add a bash user-data script

Downloads files from S3

PART 4 — S3 Bucket & AWS CLI Usage
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✅ Create an S3 bucket

✅ Upload images/config files using AWS CLI

✅ Configure bucket policy (optional)

✅ Enable versioning

PART 5 — Elastic Load Balancer (ELB)
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Create an Application Load Balancer (ALB):

- ✅ Target group
- ✅ Health checks
- ✅ Register EC2 instances

PART 6 — Auto Scaling Group (ASG)
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Create an Auto Scaling Group for high availability:

- ✅ Configure launch template
- ✅ Min size: 1
- ✅ Max size: 4
- ✅ Scaling policy based on CPU usage

PART 7 — EFS Setup (Shared Storage)
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Create EFS and mount to EC2.

Use shared directory for:

- ✅ Shared HTML content
- ✅ Application logs

PART 8 — RDS MySQL in Private Subnets
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Students deploy:

- ✅ MySQL RDS
- ✅ Multi-AZ
- ✅ Password-based authentication
- ✅ Connect EC2 → RDS using MySQL client

PART 9 — Test High Availability
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Students must:

✅ Stop an EC2 instance

✅ Observe ALB routing traffic

✅ Increase traffic and see auto scaling

✅ Validate EFS shared data remains available

✅ Validate EC2 can read/write to RDS

THIS TASK COVERS ALL REQUIRED SERVICES
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Topic	**Covered in Task**
Linux	Bash, Apache setup, EBS/EFS mount
Bash	User-data automation
IAM	Roles, policies
EC2	Web server + ASG
S3	Static content, CLI
AWS CLI	S3, EC2, IAM
ELB	Application Load Balancer
Auto Scaling	Launch Template, scale policies
VPC	Full custom network
EBS	Mounted volume
EFS	Shared storage
RDS	MySQL database